Final Report – Human-Centered AI

Prototype Design

CREATING MEANINGFUL EXPERIENCES

Applied Data Science & AI

Date

Breda University OF APPLIED SCIENCES

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Task 4.1: Create a Wireframe Prototype

Include the link to your (wireframe) prototype. Make sure it is testable.

Explain your design choices, emphasizing how they incorporate human-centered AI design principles.

https://www.figma.com/proto/arqB2CTn4xUDhsoHehimRg/Mashtal-for-plant-care--Community-?node-id=110-18&t=6kUdWy20Hbud19He-1

Design Choices and Human-Centered AI Principles:

1. **Ease of Navigation**:
    - The wireframe is designed with **clear, simple navigation** to allow users to easily access essential features such as plant health diagnosis, care instructions, and community sharing. By simplifying navigation, I reduce the cognitive load on users, ensuring the application remains intuitive and user-friendly for plant owners.
2. **User-Centric Interface**:
    - The layout emphasizes **large buttons** and **clear labels** for key actions like uploading an image of a plant and receiving care recommendations. This design choice aligns with human-centered AI principles by ensuring that the application is intuitive, reducing the friction for users who may not be tech-savvy.
3. **Accessibility**:
    - Colors and font sizes are chosen to enhance **readability** and ensure accessibility, particularly for users with varying degrees of visual ability. The high contrast between text and background enhances visibility, making it easier for users to read the content, which is an important principle in human-centered design.
4. **Responsive Feedback**:
    - The wireframe includes sections for **real-time feedback**, such as plant care recommendations based on an image upload. This feedback is crucial for users, especially plant owners, to take immediate action on their plant's health. Real-time feedback ensures that users feel informed and confident in their plant care decisions.
5. **Human-Centered AI Integration**:
    - The design incorporates **AI-based diagnoses** based on uploaded images of plants, ensuring that the AI system provides accurate and relevant care instructions. This leverages human-centered AI by ensuring that the model's predictions are aligned with the user's needs and expectations, ultimately enhancing the plant care experience.

Task 4.2: Think-Aloud Study

A. Method

1. For this think-aloud study, I selected three participants with varying levels of familiarity with plant care applications. The aim was to evaluate how intuitive and accessible the app design is, especially for non-expert users.
2. Participants were given a short persona to help them imagine a realistic use case. Then, they were asked to complete a set of tasks using the wireframe prototype while verbalizing their thoughts. The tasks reflected a typical user journey and were chosen to assess the core interactions within the app:
3. Tap the screen to start
4. Log in to the account
5. Tap the Scan icon
6. Take a photo of the plant
7. Go back to the previous screen

8. Tap the Home button
9. After completing the tasks, participants were asked four open-ended feedback questions to gather qualitative insights about their experience:
1. How was your experience using the app?
2. Was there any part of the process that felt confusing or difficult?
3. Overall, how did you find the app in terms of usability and design?
4. Is there anything you would change or add to improve the experience?

B. Data

Add a link to the transcriptions of the interviews and your personal notes.

feedback3.mp4
feedback2.mp4
feedback1.mp4

Personal Notes
During the think-aloud study, the participant who recently started caring for house plants found the app visually appealing due to its minimalistic design, clear icons, and readable fonts. They appreciated the straightforward process of scanning plants and navigating through the app, emphasizing ease of use as a key strength.
The participant reported no confusion or difficulties during the interaction, highlighting intuitive navigation between the scan, home, and profile screens. The app's simple layout was seen as suitable for everyday use by plant enthusiasts who want to regularly monitor their plants' health.
However, the participant suggested a need for better clarity regarding the app's business goals and overall purpose, indicating that this could enhance user understanding and engagement.
An additional feature recommendation was to allow users to manually upload photos to help train the AI system. This would empower users to contribute to improving the app's accuracy and personalization.
Overall, this feedback confirms the app's usability strengths while providing valuable insights on potential improvements to increase user involvement and clarity about the app's objectives.

C. Results

C. Results
From the thematic analysis of the think-aloud study, several recurring themes emerged:

1. **User Confusion with the Login Process**:
   - Multiple participants found the login process slightly confusing, especially when it wasn't clear whether the app required creating an account or simply logging in. One participant commented: *"I wasn't sure if I needed to make an account right away or if I could skip this step."*
2. **Ease of Use with the Scanning Feature**:
   - Participants enjoyed the ease of scanning their plant, but a few mentioned they weren't sure when the photo had been successfully taken. One participant stated: *"I'm not sure if the app has captured the image properly. Is there any confirmation?"*
3. **Navigation Concerns**:
   - Participants experienced confusion with going back to the previous screen and the home button. One user mentioned: *"I feel like I have to do extra steps to go back, and I'm not sure how to return to the home screen smoothly."*
4. **Overall Usability**:
   - Most users found the app easy to use, and they appreciated the simple layout. However, some felt that the lack of feedback on certain actions made the experience less intuitive.

D. Discussion

Describe how you have used the findings of this study to either validate or improve your application design.
The findings of the study have provided valuable insights into the areas of improvement for the app design:

1. **Improving the Login Process**:
    - Based on the feedback, I will make the login process more intuitive by adding clearer instructions about the account creation process. A potential solution is to provide an option to log in with social media accounts or Google to simplify the process.
2. **Improving Feedback on Scanning**:
    - I will add a confirmation message or animation when the app successfully captures a photo of the plant to reassure users. This small change can improve the overall experience, particularly for first-time users.
3. **Enhancing Navigation**:
    - The navigation experience will be improved by adding clear labels and more intuitive back and home buttons, which could prevent confusion during navigation.
4. **User Interface Adjustments**:
    - Based on usability feedback, I will refine the app layout by testing different designs for the scanning screen, ensuring that users get proper visual feedback after taking a photo.

Task 4.3: A/B Testing

Method

To evaluate the impact of visual design changes on user experience, we conducted an A/B test using two wireframe prototypes of our plant identification app:

- **Version A (Baseline):** A clean, neutral-themed design with standard button colors and layout.
- **Version B (Modified):** Identical in layout and functionality, but featuring a different color scheme and updated button styling to assess whether such visual changes influence perceived usability.

Participants were randomly assigned to interact with either Version A or Version B. After using the app, each participant completed a **Likert-scale survey** (1 = Completely Disagree, 7 = Completely Agree) evaluating:

1. I understood what I could use the app for.
2. I found the application intuitive to use.
3. I thought the application was useful.
4. I enjoyed using the application.
5. The buttons in the app were self-explanatory.

B. Data

[Link to Cleaned Data] FOR A
[Link to Cleaned Data] FOR B
Assumption Test Results

VERSION A TEST

VERSION B TEST

C. Results

- **Assumption Testing:**
- **Normality:** None of the five survey questions met the normality assumption (Shapiro-Wilk p-values < 0.05).
- **Variance Homogeneity:** All questions except Question 1 met the assumption of equal variances (Levene's test p > 0.05).

Given these assumption violations, we used **bootstrapped independent samples t-tests** to compare responses between Version A and Version B.

- **Findings:**

All **five questions showed statistically significant differences**, with **Version B scoring higher** in every case. This suggests that even small visual changes (such as button colors and theme updates) had a meaningful impact on how users perceived the app.

[Link to T-test Notebook]

D. Discussion

Contrary to initial expectations, the updated visual design in **Version B led to significant improvements in perceived usability and user satisfaction**. While functionality remained unchanged between versions, changes in **theme and button styling** resulted in consistently higher user ratings across all measured aspects.

- **Design Implications:**

1. **Visual design matters:** Even minimal changes in UI styling—such as color scheme and button aesthetics—can significantly influence how users interpret and engage with an app.
2. **Version B as a new baseline:** Given the consistent positive results, Version B will be adopted as the foundation for future design iterations.
3. **User-centered refinements:** Going forward, we will continue refining the interface by combining **visual appeal with usability enhancements**, ensuring that all updates are validated through both quantitative testing and user feedback.

These results reinforce the importance of **evidence-based design decisions**. Visual styling is not just cosmetic—it can meaningfully shape user experience when executed thoughtfully.

Task 4.4: Create a Final Demo Video

Add a link to the demo video here.
https://github.com/BredaUniversityADSAI/2024-25c-fai1-adsai-CerenKurt243910/blob/main/Deliverables/Recording%202025-04-11%20050531.mp4

References

List the references used in the report here.

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